第7章　动态电路的暂态分析

(dynamic circuit) (transient analysis)

7.1 动态电路概述

7.2 电路的初始条件

7.3 一阶电路的暂态响应

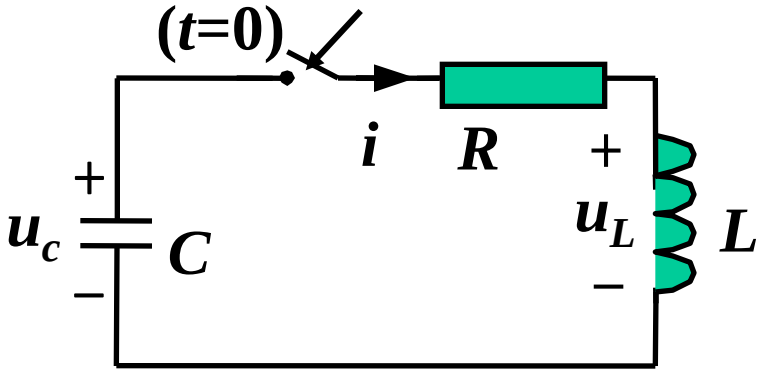
7.4 一阶电路的阶跃和冲激响应

7.5 二阶电路的响应

7.6 高阶电路过渡过程的求解方法

7.5 二阶电路的响应

7.5.1 零输入响应



RLC串联电路的零输入响应

已知 $u_C(0^-)=U_0$ $i(0^-)=0$

求 $u_C(t)$, $i(t)$, $u_L(t)$.

解 $Ri+u_L-u_C=0$ $i=-C\dfrac{\mathrm{d}u_C}{\mathrm{d}t}$ $u_L=L\dfrac{\mathrm{d}i}{\mathrm{d}t}=-LC\dfrac{\mathrm{d}^2u_C}{\mathrm{d}t^2}$

$$LC\frac{\mathrm{d}^2u_C}{\mathrm{d}t^2}+RC\frac{\mathrm{d}u_C}{\mathrm{d}t}+u_C=0$$

特征方程为 $LCP^2+RCP+1=0$

$$P_{1,2}=\frac{-RC\pm\sqrt{R^2C^2-4LC}}{2LC}=-\frac{R}{2L}\pm\sqrt{(\frac{R}{2L})^2-\frac{1}{LC}}$$

$$p_{1,2}=-\frac{R}{2L}\pm\sqrt{\left(\frac{R}{2L}\right)^2-\frac{1}{LC}}$$

$$P_{12}=-\alpha\pm j\omega_d=-\alpha\pm j\sqrt{\omega_0{}^2-\alpha^2}$$

根的性质不同，响应的变化规律也不同

$R>2\sqrt{\frac{L}{C}}$ 二个不等负实根 $u_C=A_1e^{p_1t}+A_2e^{p_2t}$

$R=2\sqrt{\frac{L}{C}}$ 二个相等负实根 $u_C=(A_1+A_2t)e^{pt}$

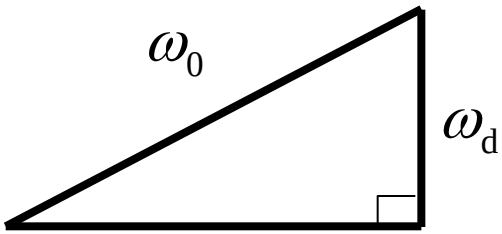
$R<2\sqrt{\frac{L}{C}}$ 一对共轭复根 $u_C=A_1e^{p_1t}+A_2e^{p_2t}$

$P_{12}=-\alpha\pm j\omega_d$ $=Ke^{-\alpha t}\sin(\omega_d t+\beta)$

衰减系数 $\alpha=\frac{R}{2L}$

自由振荡角频率/自然角频率 $\omega_0=\frac{1}{\sqrt{LC}}$

衰减振荡角频率 $\omega_d=\sqrt{\omega_0^2-\alpha^2}$

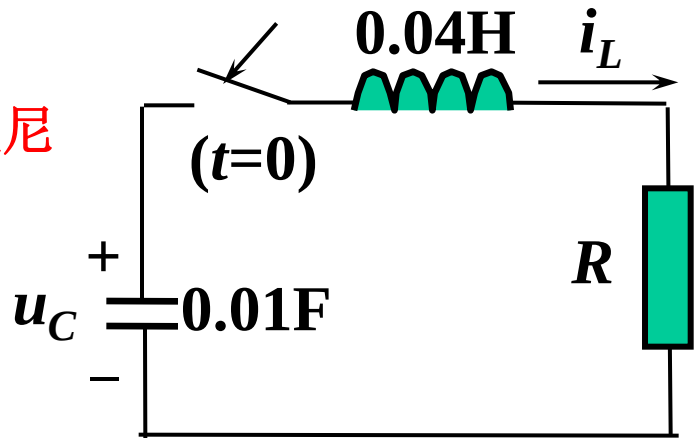


$$\omega_0^2=\omega_d^2+\alpha^2$$

$$\frac{\mathrm{d}^2u_C}{\mathrm{d}t^2}+\frac{R}{L}\frac{\mathrm{d}u_C}{\mathrm{d}t}+\frac{1}{LC}u_C=0$$

$$b^2-4ac=625R^2-10000$$

例：R分别为5Ω、4Ω、1Ω、0Ω时求$u_C(t)$、$i_L(t)$，$t\geq 0$。



$u_C(0^-)=3\mathrm{V}$

$i_L(0^-)=0$

$R=5\,\Omega$ →

$b^2-4ac=5625>0$

$p_1=-25 \quad p_2=-100$ 过阻尼

$u_C(t)=A_1\mathrm{e}^{-25t}+A_2\,\mathrm{e}^{-100t}$

$R=4\,\Omega$ →

$b^2-4ac=0$ 临界阻尼

$P_1=P_2=-50$

$u_C(t)=A_1\mathrm{e}^{-50t}+A_2t\,\mathrm{e}^{-50t}$

$R=1\,\Omega$ →

$b^2-4ac=-9375<0$ 欠阻尼

$p_{1,2}=-12.5\pm\mathrm{j}48.4$

$u_C(t)=K\mathrm{e}^{-12.5t}\sin(48.4t+\theta)$

3. 用初值确定待定系数

$u_C(0)=3\text{V}$

$\left.\dfrac{\mathrm{d}u_C}{\mathrm{d}t}\right|_{t=0^+}=0$

$R=5\Omega$

$$u_C(t)=A_1\mathrm{e}^{-25t}+A_2\,\mathrm{e}^{-100t}$$

$$\begin{cases}A_1+A_2=3\\ -25A_1-100A_2=0\end{cases}\Rightarrow A_1=4\quad A_2=-1$$

$$u_C(t)=4\mathrm{e}^{-25t}-\mathrm{e}^{-100t}\text{V}\quad(t\geq 0)$$

$R=4\Omega$

$$u_C(t)=A_1\mathrm{e}^{-50t}+A_2t\,\mathrm{e}^{-50t}$$

$$\begin{cases}A_1=3\\ -50A_1+A_2=0\end{cases}\Rightarrow A_1=3,\ \ A_2=150$$

$$u_C(t)=3\mathrm{e}^{-50t}(1+50t)\text{V}\quad(t\geq 0)$$

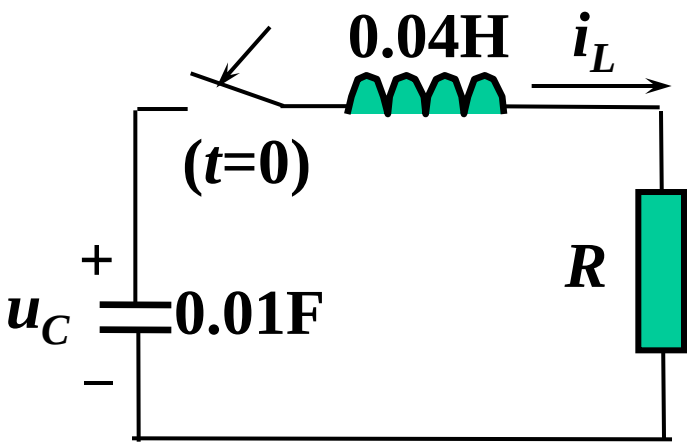
$R=1\Omega$

$$u_C(t)=K\mathrm{e}^{-12.5t}\sin(48.4t+\theta)$$

$$\begin{cases}K\sin\theta=3\\ -12.5K\sin\theta+48.4K\cos\theta=0\end{cases}\Rightarrow K=3.1,\ \ \theta=75.5^{\circ}$$

$$u_C(t)=3.10\mathrm{e}^{-12.5t}\sin(48.4t+75.5^{\circ})\text{V}\quad(t\geq 0)$$

$$C\frac{\mathrm{d}u_C}{\mathrm{d}t}=-i_L$$



$R=5\Omega$

$$\begin{cases} u_C(t)=4\mathrm{e}^{-25t}-\mathrm{e}^{-100t}\mathrm{V} \quad (t\geq 0) \\ i(t)=\mathrm{e}^{-25t}-\mathrm{e}^{-100t}\mathrm{A} \quad (t\geq 0) \end{cases}$$

$R=4\Omega$

$$\begin{cases} u_C(t)=3\mathrm{e}^{-50t}(1+50t)\mathrm{V} \quad (t\geq 0) \\ i(t)=75t\mathrm{e}^{-50t}\mathrm{A} \quad (t\geq 0) \end{cases}$$

$R=1\Omega$

$$\begin{cases} u_C(t)=3.10\mathrm{e}^{-12.5t}\sin(48.4t+75.5^{\mathrm{o}})\mathrm{V} \quad (t\geq 0) \\ i(t)=1.55\mathrm{e}^{-12.5t}\sin 48.4t\ \mathrm{A} \quad (t\geq 0) \end{cases}$$

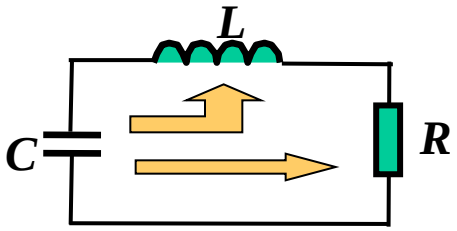
4. 波形与能量传递

$R=5\Omega$ 过阻尼，无振荡放电

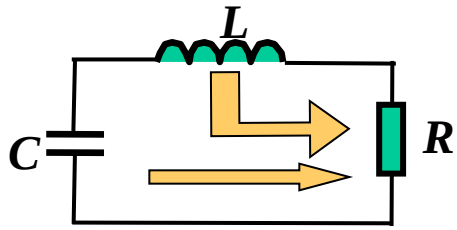
$u_C(t)=4e^{-25t}-e^{-100t}V \quad (t\geq 0)$

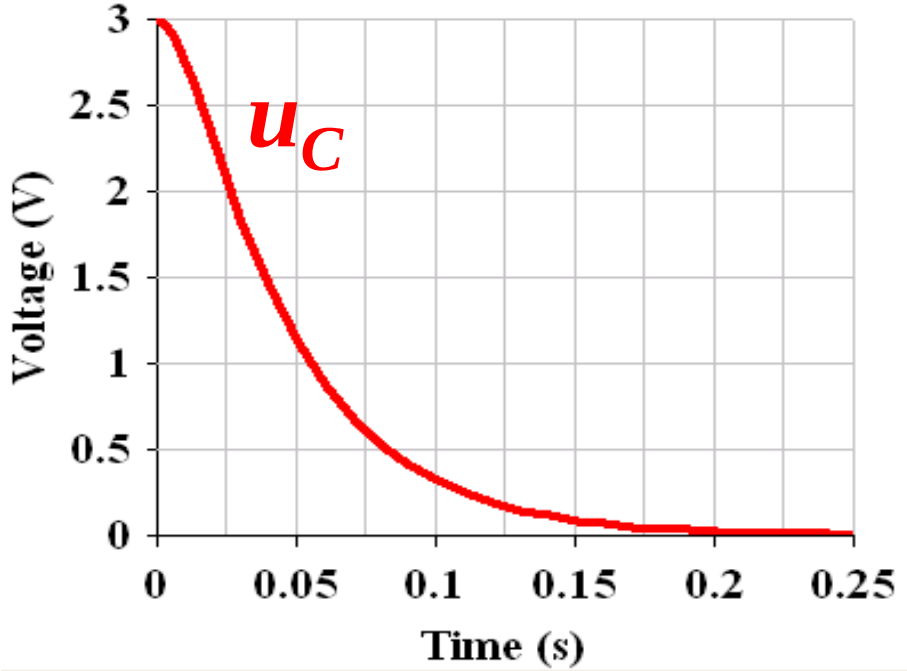
$i(t)=e^{-25t}-e^{-100t}A \quad (t\geq 0)$

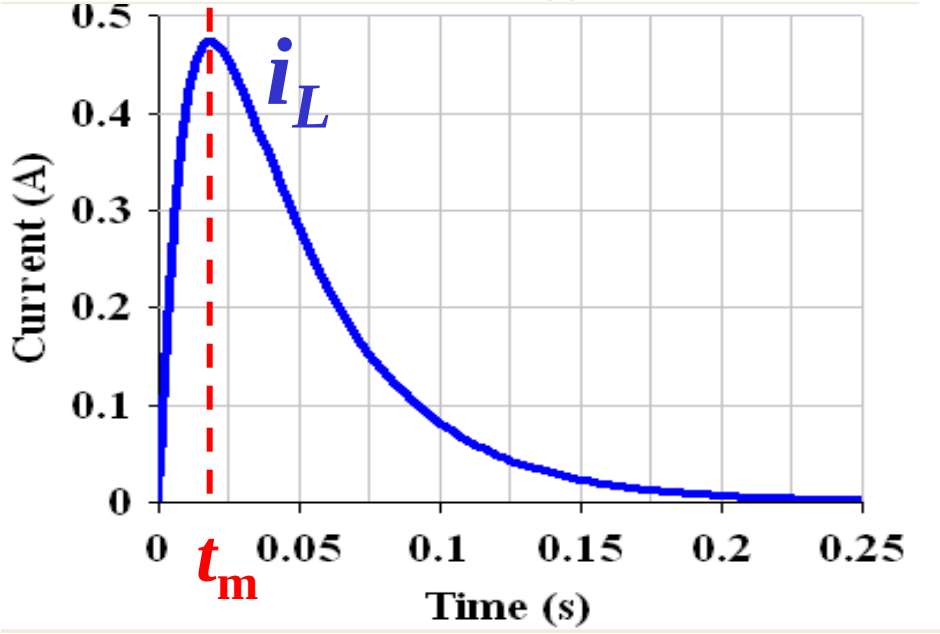
$0<t<t_m$ u_C 减小, i 增加.



$t>t_m$ u_C 减小 , i 减小 。





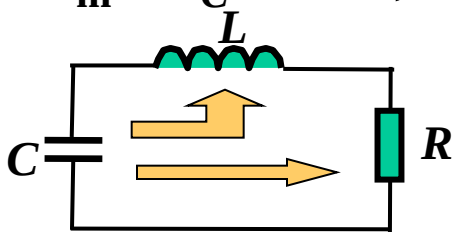


$R=4\Omega$ 临界阻尼，无振荡放电

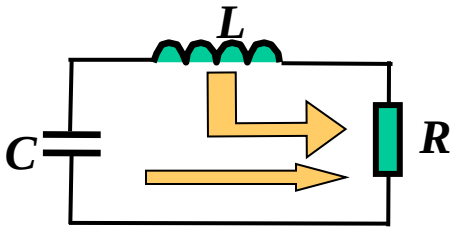
$u_C(t)=3\mathrm{e}^{-50t}(1+50t)\mathrm{V}\quad(t\geq 0)$

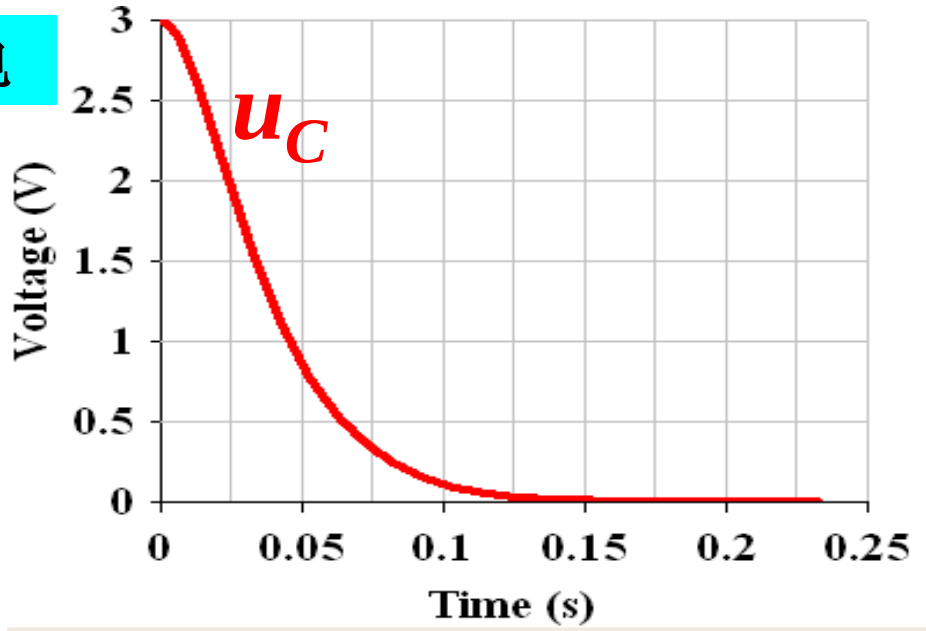
$i(t)=75t\mathrm{e}^{-50t}\mathrm{A}\quad(t\geq 0)$

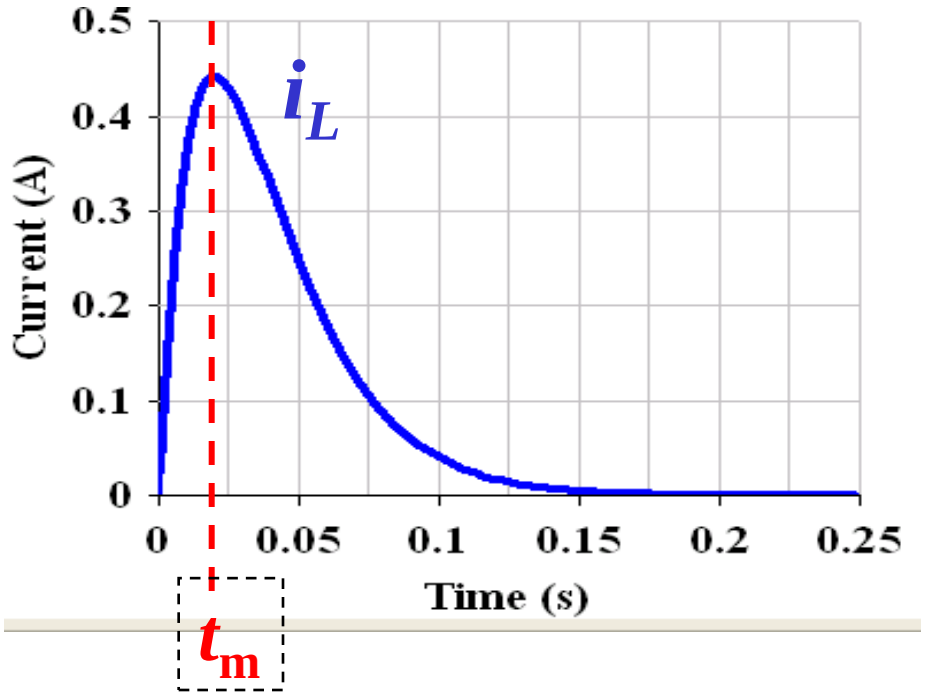
$0<t<t_m$ u_C 减小，i 增加.



$t>t_m$ u_C 减小，i 减小.



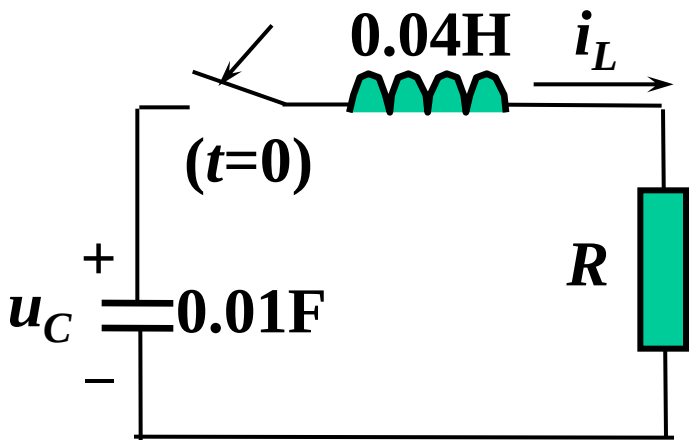


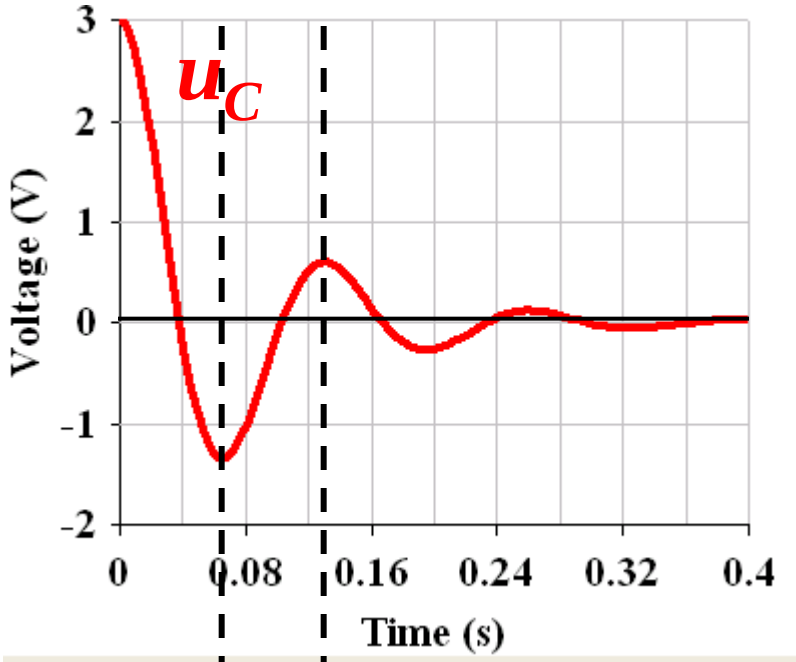


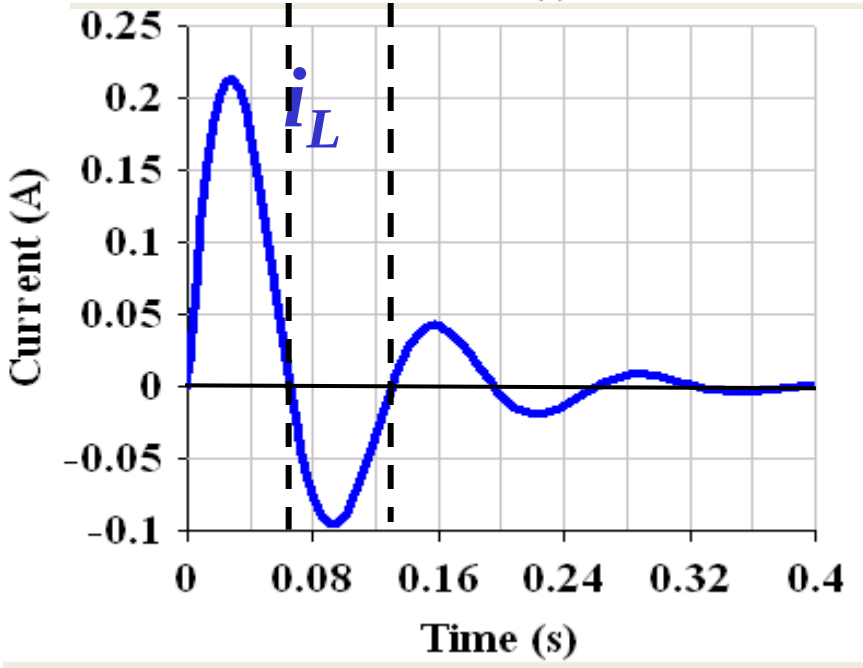
$R=1\Omega$ 欠阻尼，振荡放电

$$u_C(t)=3.10e^{-12.5t}\sin(48.4t+75.5^{o})V \quad (t\geq 0)$$

$$i(t)=1.55e^{-12.5t}\sin 48.4t\ A \quad (t\geq 0)$$



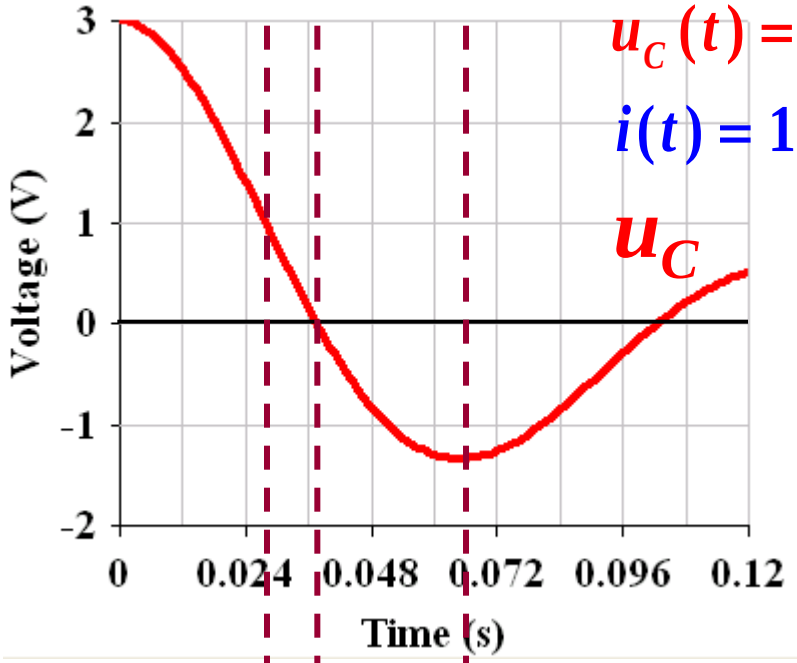


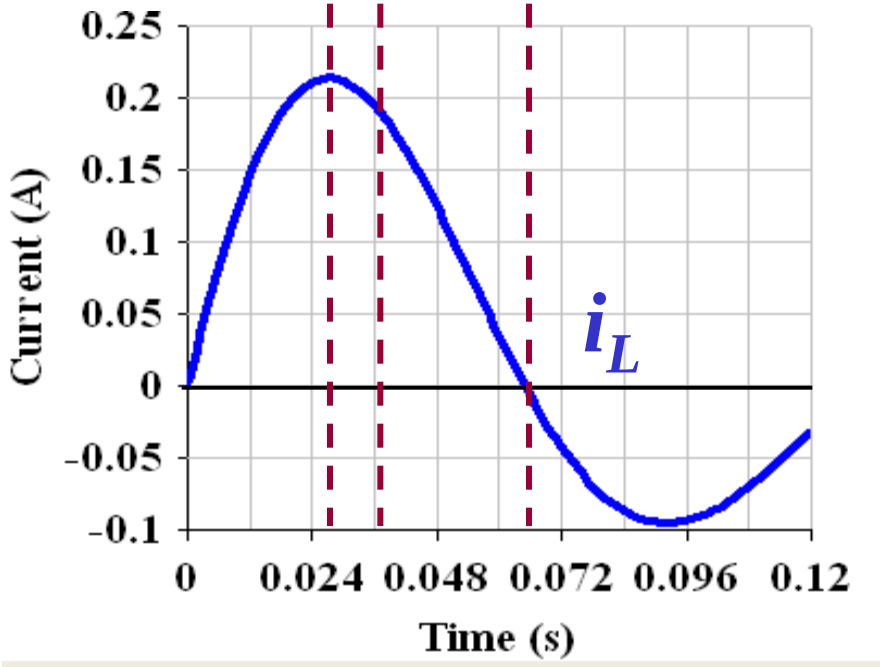


讨论半个周期中能量的关系

$u_C(t) = 3.10e^{-12.5t}\sin(48.4t + 75.5^o)V$

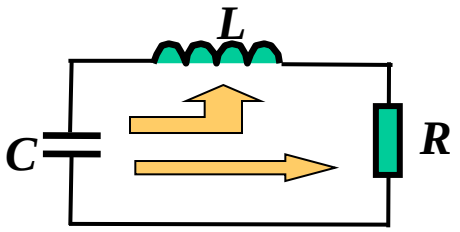
$i(t) = 1.55e^{-12.5t}\sin 48.4t\ A \quad (t \geq 0)$





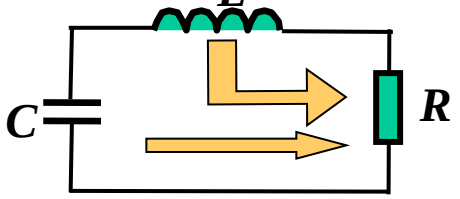
$0 \leq 48.4t \leq 90^o$

u_C 减小， i 增加



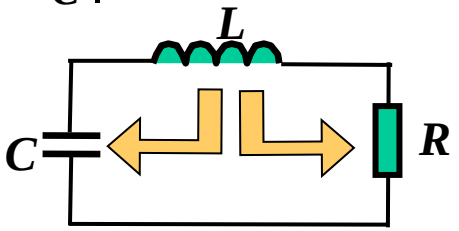
$90^o \leq 48.4t \leq 104.5^o$

u_C 减小，i 减小



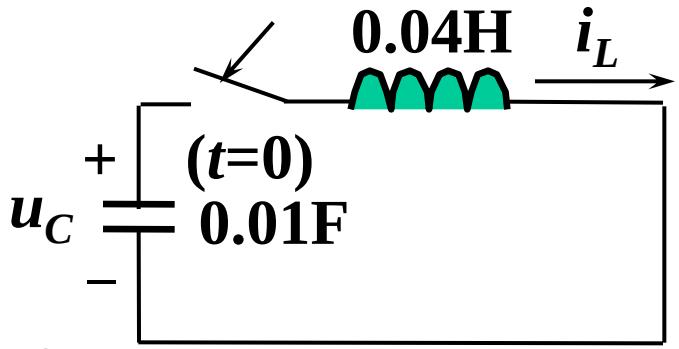
$104.5^o \leq 48.4t \leq 180^o$

$|u_C|$ 增加，i 减小



周而复始，电阻不断消耗能量，u_C i_L 衰减到零。

$R=0$ 无阻尼振荡



$$LC\frac{\mathrm{d}^2u_C}{\mathrm{d}t}+u_C=0$$

$$p^2+2500=0 \qquad p=\pm\mathrm{j}50$$

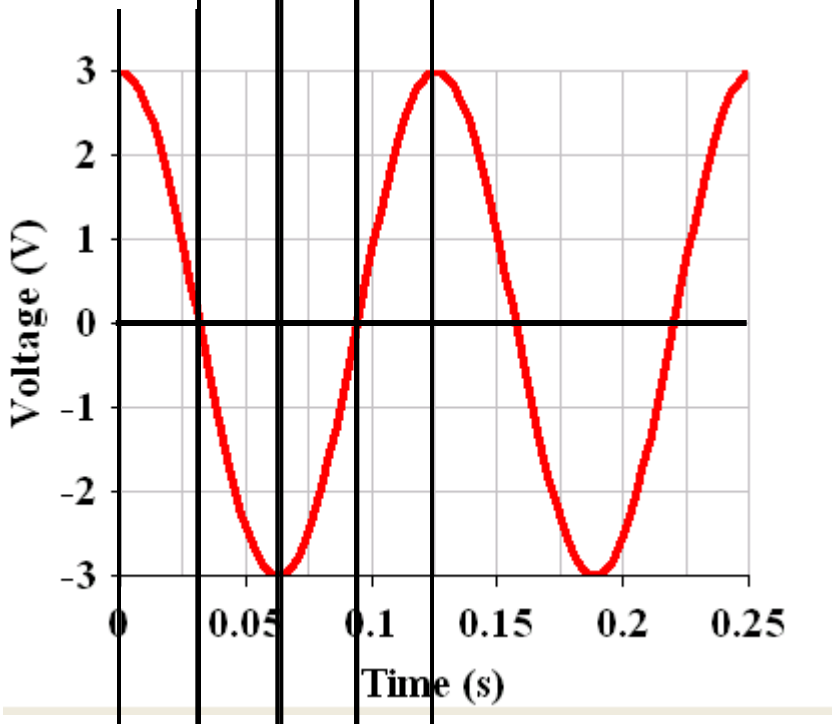
$$u_C(t)=K\sin(50t+\theta)$$

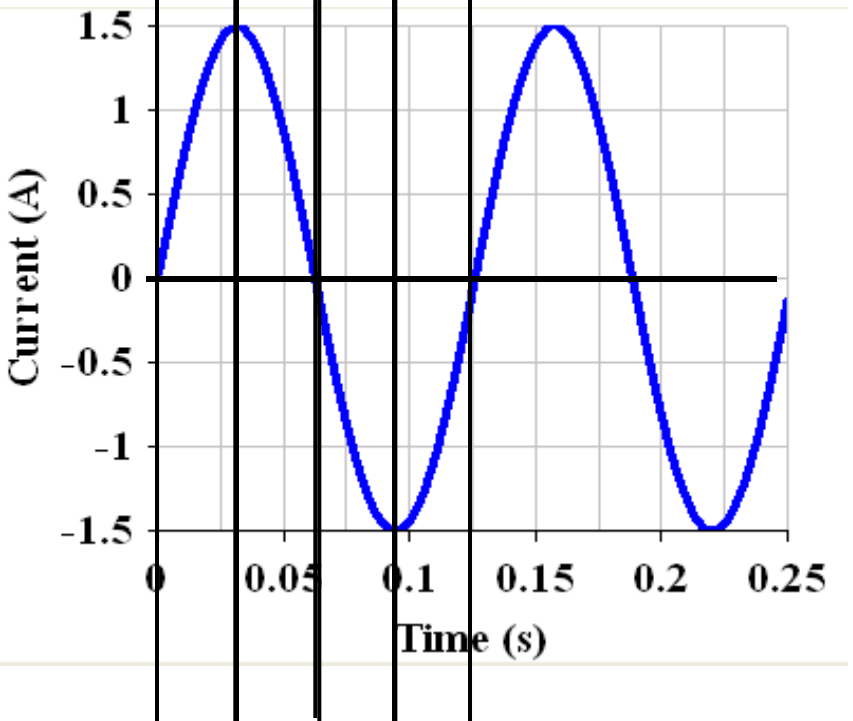
$$u_C(0)=3,\quad \left.\frac{\mathrm{d}u_C}{\mathrm{d}t}\right|_{t=0^+}=0$$

$$K=3,\quad \theta=90^{\circ}$$

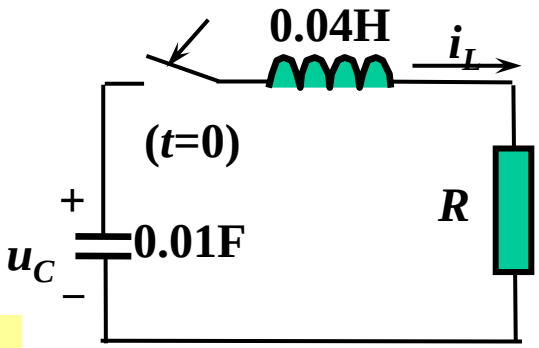
$$u_C(t)=3\cos 50t\ \mathrm{V}\quad (t\geq 0)$$

$$i(t)=1.5\sin 50t\ \mathrm{A}\quad (t\geq 0)$$





二、用直觉解法定性画支路量的变化曲线



1. 过阻尼或临界阻尼（无振荡衰减）

以过阻尼为例。

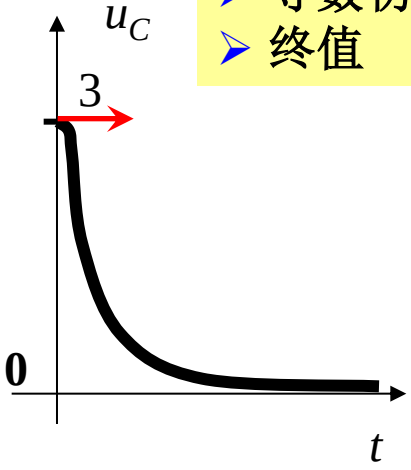
- 初值
- 导数初值
- 终值

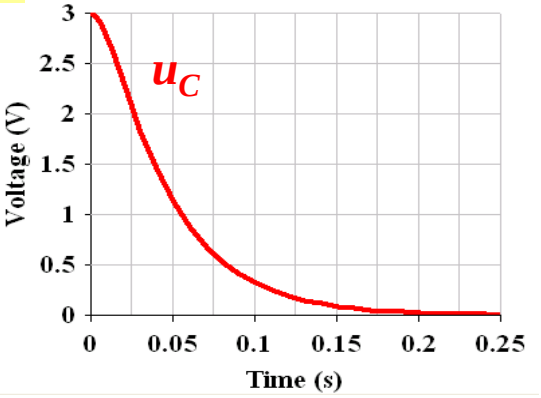
$u_C(0^-) = 3\text{V}$

$i_L(0^-) = 0$

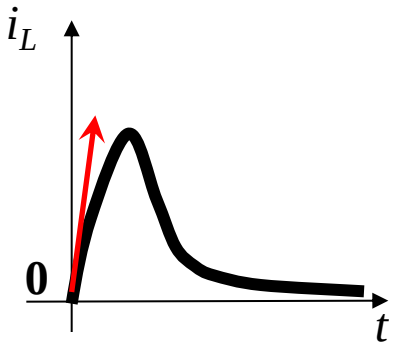
$$p_1 = -25, \quad p_2 = -100$$

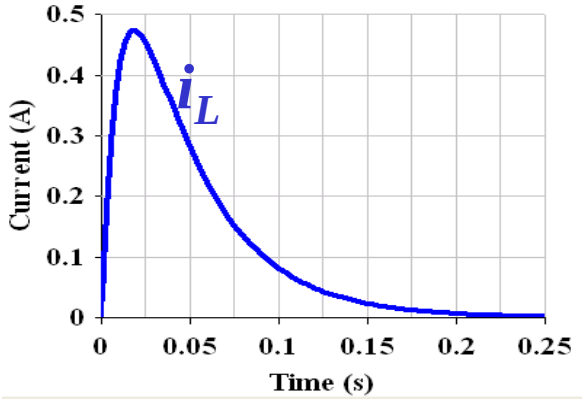
$$\begin{cases} u_C(0^+) = 3\text{V} \\ \left.\dfrac{\mathrm{d}u_C}{\mathrm{d}t}\right|_{t=0^+} = -\dfrac{1}{C} i_L(0^+) = 0 \end{cases}$$





$$\begin{cases} i_L(0^+) = 0 \\ \left.\dfrac{\mathrm{d}i_L}{\mathrm{d}t}\right|_{t=0^+} = \dfrac{1}{L} u_L(0^+) = \dfrac{3}{L} \end{cases}$$



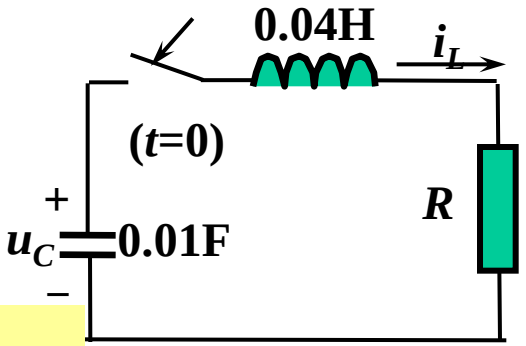


2. 欠阻尼（衰减振荡）

衰减系数α

衰减振荡角频率ω_d

$$p_{1,2} = -12.5 \pm j48.4$$



$u_C(0^-) = 3\text{V}$

$i_L(0^-) = 0$

- 初值
- 导数初值
- 终值
- 经过多少周期振荡衰减完毕

$$\begin{cases} u_C(0^+) = 3\text{V} \\ \left.\dfrac{\mathrm{d}u_C}{\mathrm{d}t}\right|_{t=0^+} = -\dfrac{1}{C} i_L(0^+) = 0 \end{cases}$$

回忆一阶电路中的时间常数τ：**3～5τ后过渡过程结束**

$$3\times\frac{1}{\alpha} = \frac{3}{12.5} = 0.24\ \text{s}$$

后过渡过程结束

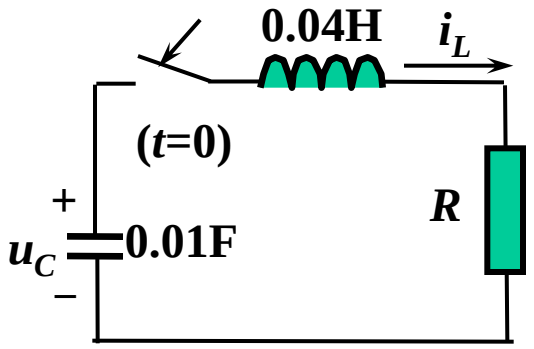
$$5\times\frac{1}{\alpha} = \frac{5}{12.5} = 0.4\ \text{s}$$

振荡周期为 $T = \dfrac{2\pi}{\omega} = \dfrac{2\pi}{48.4} = 0.13\,\text{s}$

衰减过程中有
0.24/0.13~2次振荡
或0.4/0.13≈3次振荡

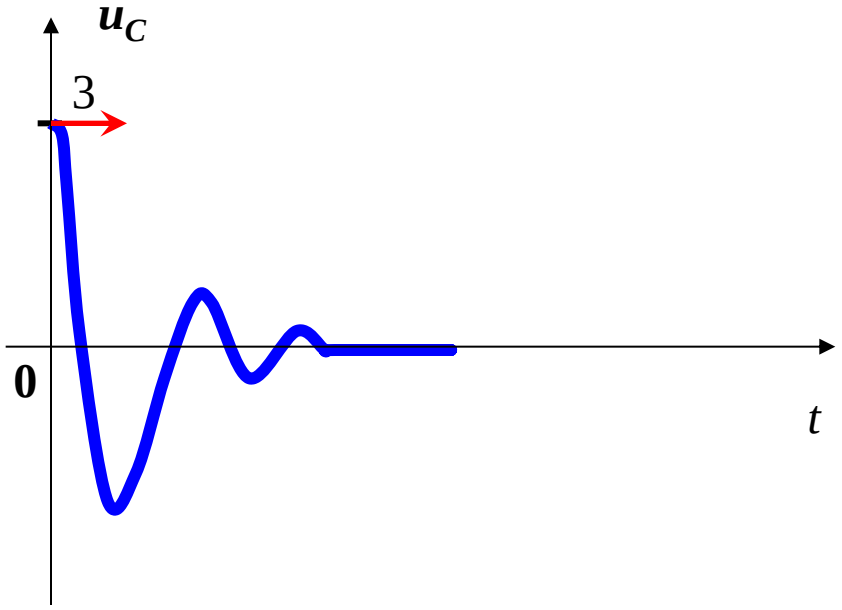
$$p_{1,2} = -12.5 \pm \mathrm{j}48.4$$

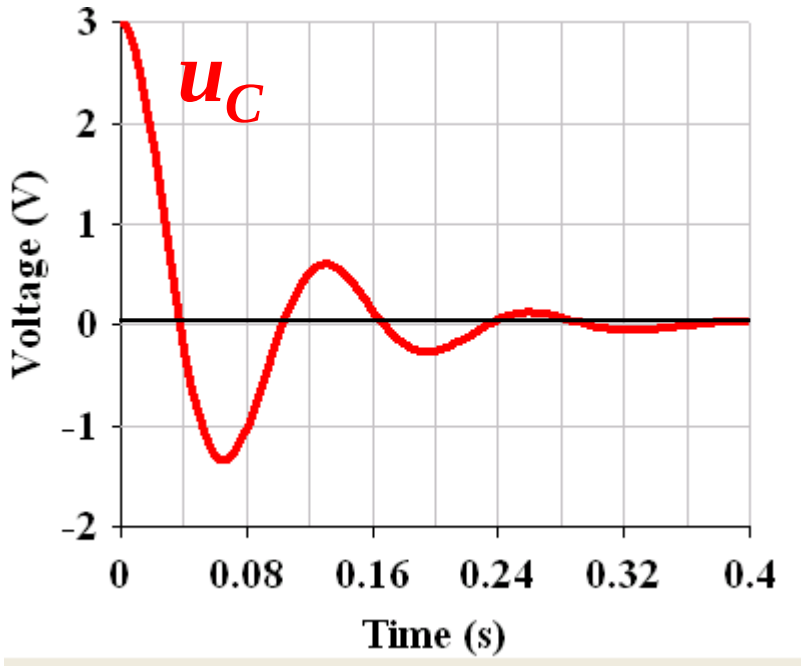
$$\begin{cases} u_C(0^+) = 3\mathrm{V} \\ \left.\dfrac{\mathrm{d}u_C}{\mathrm{d}t}\right|_{t=0^+} = -\dfrac{1}{C} i_L(0^+) = 0 \end{cases}$$



- 初值
- 导数初值
- 终值
- 经过多少周期振荡衰减完毕

衰减过程中有
0.24/0.13≈2次振荡
或0.4/0.13≈3次振荡

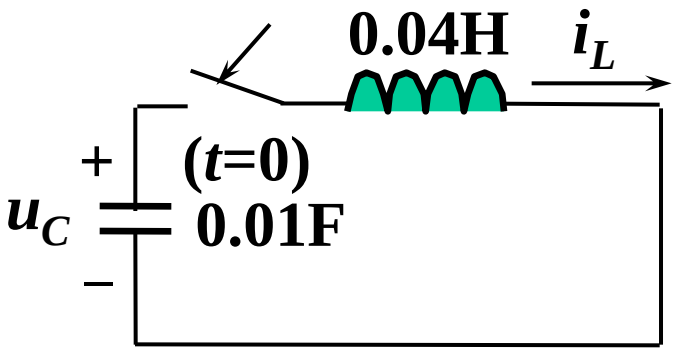




3. 无阻尼

$$p = \pm j50$$

- 初值
- 导数初值
- 最大值



$u_C(0^-) = 3V$

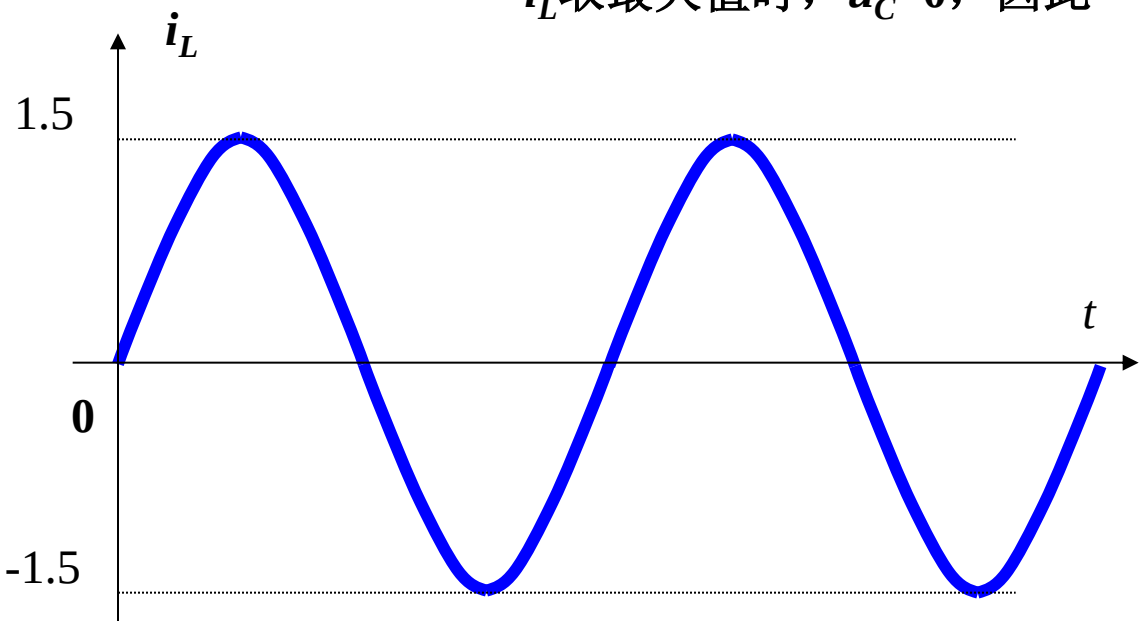
$i_L(0^-) = 0$

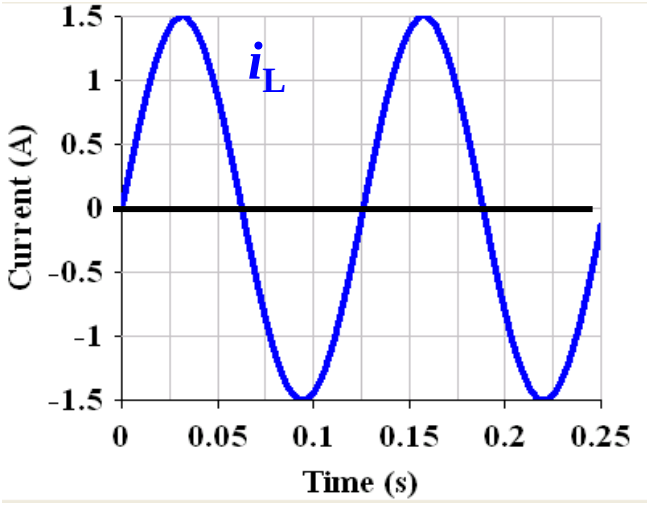
$$\begin{cases} i_L(0^+) = 0 \\ \left.\dfrac{di_L}{dt}\right|_{t=0^+} = \dfrac{1}{L}u_L(0^+) = \dfrac{3}{L} \end{cases}$$

因为无阻尼，所以能量守恒

$$\frac{1}{2}Cu_C^2(0) + \frac{1}{2}Li_L^2(0) = \frac{1}{2}Cu_C^2(t) + \frac{1}{2}Li_L^2(t)$$

i_L取最大值时，u_C=0，因此 $i_{Lmax} = \sqrt{\dfrac{C}{L}}u_C(0) = 1.5A$



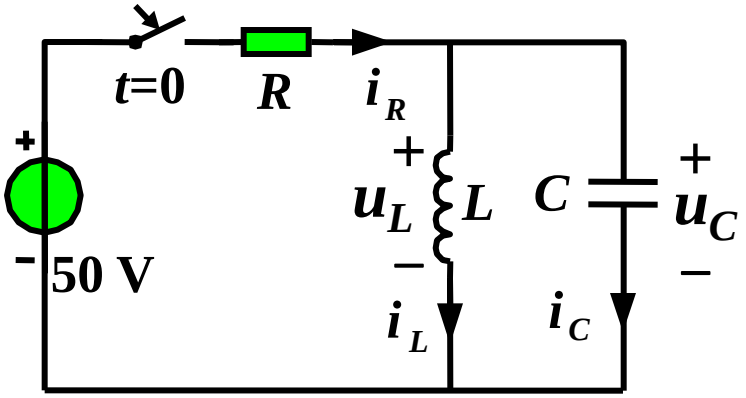


7.5.2 全响应—5电路计算

已知： $i_L(0)=2\text{A}$ $u_C(0)=0$

$R=50\Omega$, $L=0.5\text{H}$, $C=100\mu\text{F}$

求： $i_L(t)$, $i_R(t)$ 。



解 (1) 列微分方程——电路1

$$\frac{50-L\dfrac{\mathrm{d}i_L}{\mathrm{d}t}}{R}=i_L+C\frac{\mathrm{d}u_C}{\mathrm{d}t}$$

$$u_C=u_L=L\frac{\mathrm{d}i_L}{\mathrm{d}t}$$

$$RLC\frac{\mathrm{d}^2i_L}{\mathrm{d}t^2}+L\frac{\mathrm{d}i_L}{\mathrm{d}t}+Ri_L=50$$

$$\frac{\mathrm{d}^2i_L}{\mathrm{d}t^2}+200\frac{\mathrm{d}i_L}{\mathrm{d}t}+2\times10^4i_L=2\times10^4$$

$$\frac{\mathrm{d}^2 i_L}{\mathrm{d}t^2} + 200\frac{\mathrm{d}i_L}{\mathrm{d}t} + 2\times10^4 i_L = 2\times10^4$$

(2)求通解(自由分量）

特征方程　　$P^2 + 200P + 20000 = 0$

特征根　　$P = -100 \pm j100$

通解　$i_L(t) = Ke^{-100t}\sin(100t + \beta)$

(3)求特解（强制分量，稳态解）—电路2

$$i_L'' = 1A$$

(4)求全解

全解　$i_L(t) = 1 + Ke^{-100t}\sin(100t + \beta)$

全解 $i_L(t)=1+Ke^{-100t}\sin(100t+\beta)$

(4)由初值定积分常数—电路3、电路4

$i_L(0^+)=2\text{A}$，$u_C(0^+)=0$ （已知）

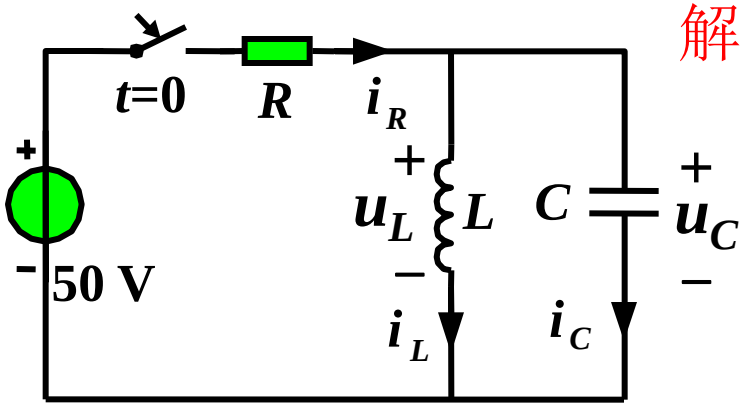
$$\frac{\mathrm{d}i_L}{\mathrm{d}t}\Big|_{0+}=\frac{1}{L}u_L(0^+)=\frac{1}{L}u_C(0^+)=0$$

$$\frac{\mathrm{d}i_L}{\mathrm{d}t}=-100Ke^{-100t}\sin(100t+\beta)+100Ke^{-100t}\cos(100t+\beta)$$

$$\begin{cases} i_L(0^+)=2 \quad \rightarrow \quad 1+K\sin\beta=2 \\ \dfrac{di_L}{dt}\Big|_{0+}=0 \quad \rightarrow \quad -100K\sin\beta+100K\cos\beta=0 \end{cases}$$

得 $K=\sqrt{2}\quad \beta=45^{\circ}$

$$\therefore i_L(t)=1+\sqrt{2}e^{-100t}\sin(100t+45^{\circ})A \quad t\geq 0$$



解1

(5) 求 $i_R(t)$ —电路5

R=50Ω
C=100μF

$$i_R = i_L + CL\frac{d}{dt}\left(\frac{di_L}{dt}\right)$$

$$i_R = \frac{1}{R}\left(50 - L\frac{di_L}{dt}\right)$$

$$i_R = \frac{1}{50}\left\{50 - 100\sqrt{2}e^{-100t}\left(-\sin(100t+45^\circ)+\cos(100t+45^\circ)\right)\right\}$$

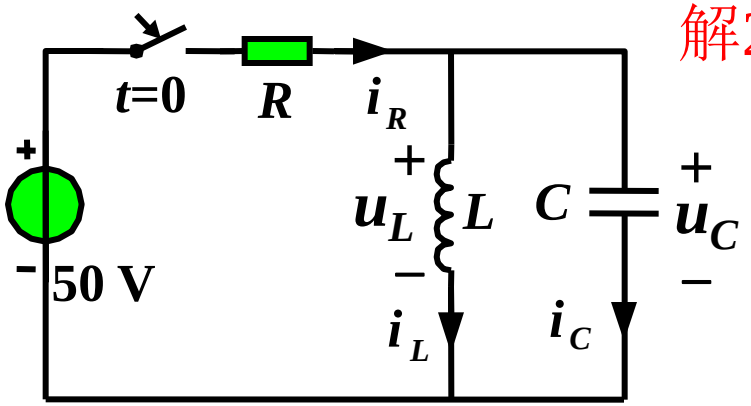
$$i_R(t) = 1 + 2e^{-100t}\sin 100t\ A \qquad t \geq 0$$

$$\therefore\ i_L(t) = 1 + \sqrt{2}e^{-100t}\sin(100t+45^\circ)A \qquad t \geq 0$$

解2

(5) 求 $i_R(t)$ —电路5

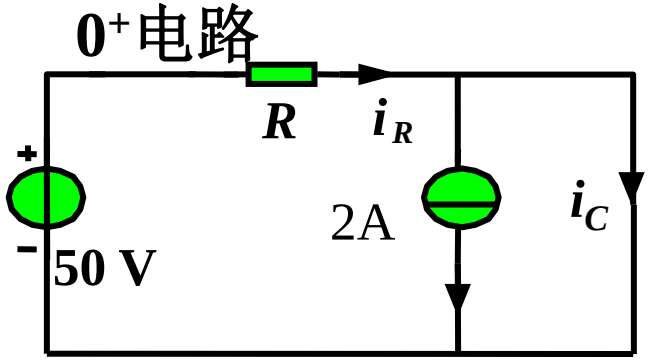
R=50Ω
C=100μF



解答形式为:

$$i_R(t)=1+Ke^{-100t}\sin(100t+\beta)$$

$$i_R(0^+)=\frac{50-u_C(0^+)}{50}=1$$



$$i_C(0^+)=-1A$$

$$\left.\frac{di_R}{dt}\right|_{0+}=\frac{d}{dt}\left(\frac{50-u_C}{R}\right)=-\frac{1}{RC}i_C(0^+)$$

$$=-\frac{-1}{50\times100\times10^{-6}}=200$$

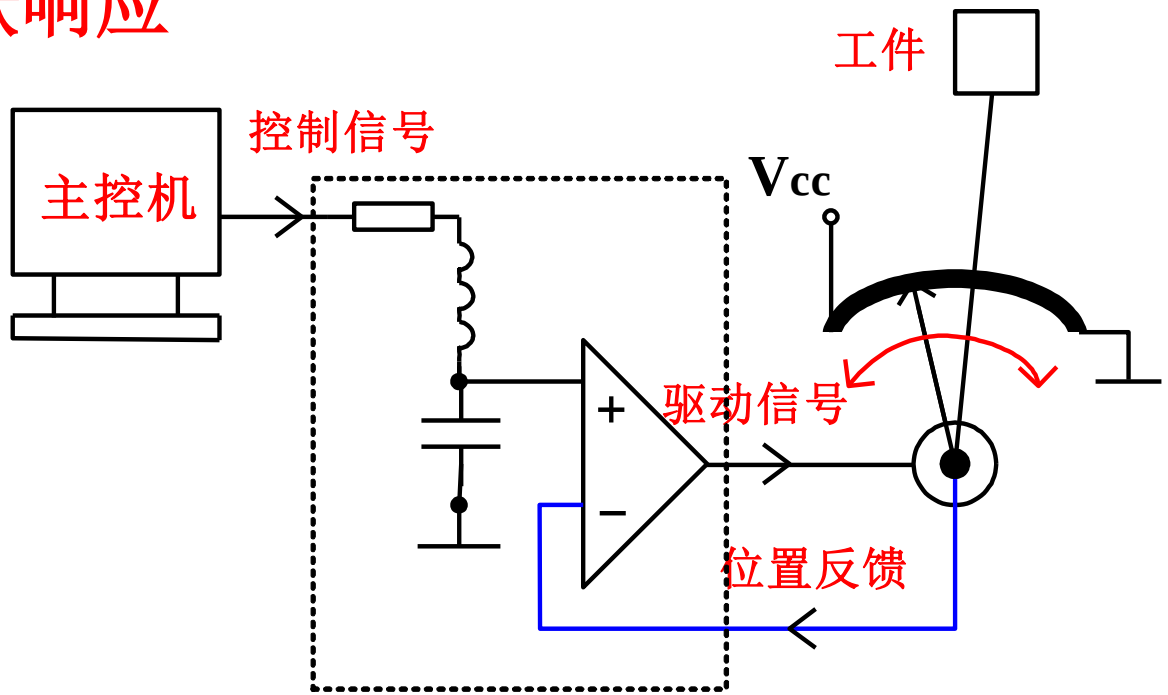
由初始值定积分常数

$$\begin{cases}1+K\sin\beta=1\\ 100K\cos\beta-100K\sin\beta=200\end{cases}\longrightarrow\begin{cases}\beta=0\\ K=2\end{cases}$$

$$i_R(t)=1+2e^{-100t}\sin100t\ A\qquad t\geq0$$

7.5.3 RLC电路阶跃响应

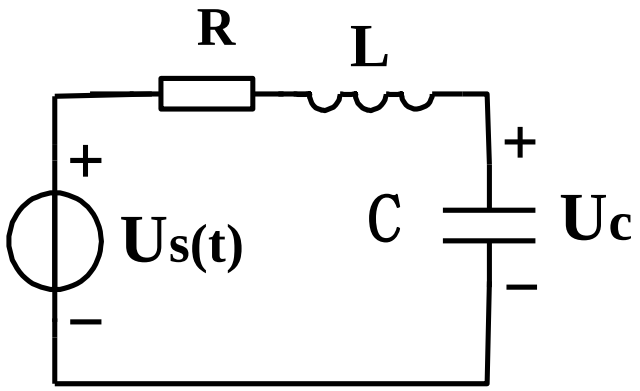
控制系统实例



数学模型：

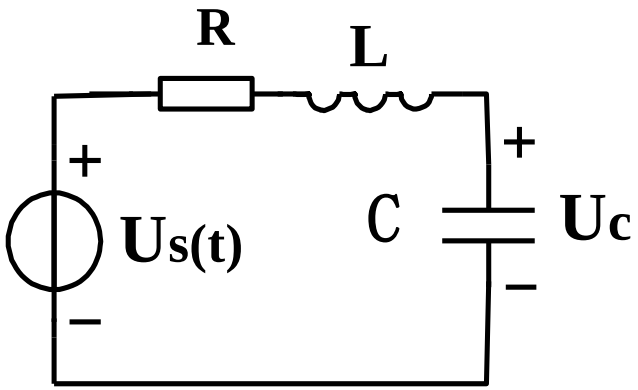
电容电压阶跃响应

$$U_S(t) = U_S \cdot 1(t)$$



例：$U_C(0^-)=0,\ \ i_L(0^-)=0.\ \ U_S(t)=U_S\cdot 1(t)$

求 $U_C(t)$



设：$L=1H,\ C=1F,\ R$ 分别为 $1\Omega, 2\Omega, 3\Omega.$

解：电路方程 $LC\dfrac{\mathrm{d}^2U_C}{\mathrm{d}t^2}+RC\dfrac{\mathrm{d}U_C}{\mathrm{d}t}+U_C=U_S$ (二阶非齐次方程).

初始条件：$U_C(0^+)=0,\ \ \left.\dfrac{dU_C}{dt}\right|_{t=0^+}=0$ 特解 (稳态解)：$U_C'=U_S$

通解：**1>.** 当 $R=3\Omega, R>2\sqrt{\dfrac{L}{C}}=2$ 过阻尼 $S_1=-0.38, S_2=-2.62$

2>.当 $R=2\Omega, R=2\sqrt{\dfrac{L}{C}}=2$ 临界阻尼 $S_1=S_2=-\dfrac{R}{2L}=-1$

3>.当 $R=1\Omega, R<2\sqrt{\dfrac{L}{C}}=2$ 欠阻尼振荡 $S_{1,2}=-\dfrac{1}{2}\pm j\dfrac{\sqrt{3}}{2}$

$$S_{1,2}=-\frac{R}{2L}\pm\sqrt{(\frac{R}{2L})^2-\frac{1}{LC}}$$

$$U_C(t)=U_S+K_1e^{-0.38t}+K_2e^{-2.62t}$$

$$U_C(t)=U_S+(A_1+A_2t)e^{-t}$$

$$U_C(t)=U_S+Ae^{-\frac{1}{2}t}\sin(\frac{\sqrt{3}}{2}t+\theta).$$

初始条件：

$$U_C(0)=0,\quad \left.\frac{\mathrm{d}U_C}{\mathrm{d}t}\right|_{t=0}=0$$

$$U_C(t)=U_S-1.17U_Se^{-0.38t}+0.17U_Se^{-2.62t}$$

$$U_C(t)=U_S-U_S(1+t)e^{-t}$$

$$U_C(t)=U_S-\frac{2}{\sqrt{3}}U_Se^{-\frac{t}{2}}\sin(\frac{\sqrt{3}}{2}t+60^\circ)$$

当 $\frac{\sqrt{3}}{2}t+\frac{\pi}{3}=\pi$ 即 $t=\frac{4\pi}{3\sqrt{3}}=2.41s$ 时，$U_C(2.41)=U_S$

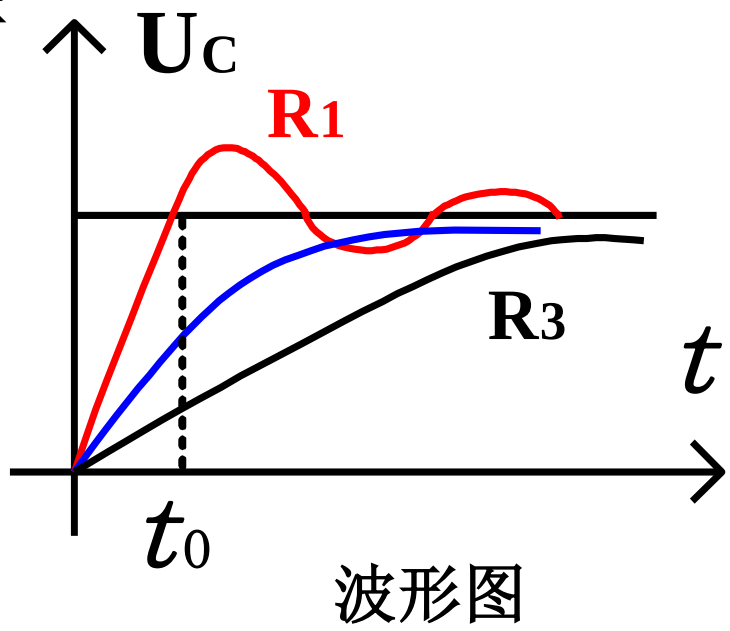
讨论: 减小**R**可使系统响应加快, 在

$t_0 = 2.41s$ 时,

$R = 1\Omega, U_C = U_S;$

$R = 2\Omega,\ U_C = 0.69U_S;$

$R = 3\Omega, U_C = 0.53U_S$.



波形图

随着**R**的增加，**Uc**的值减小，即响应速度慢，过渡过程长。

随着***R***减小, 系统出现振荡, ***R***越小, 超调量越大.

响应速度与超调量是互相关联的,在系统设计时应考虑二者之间的关系。

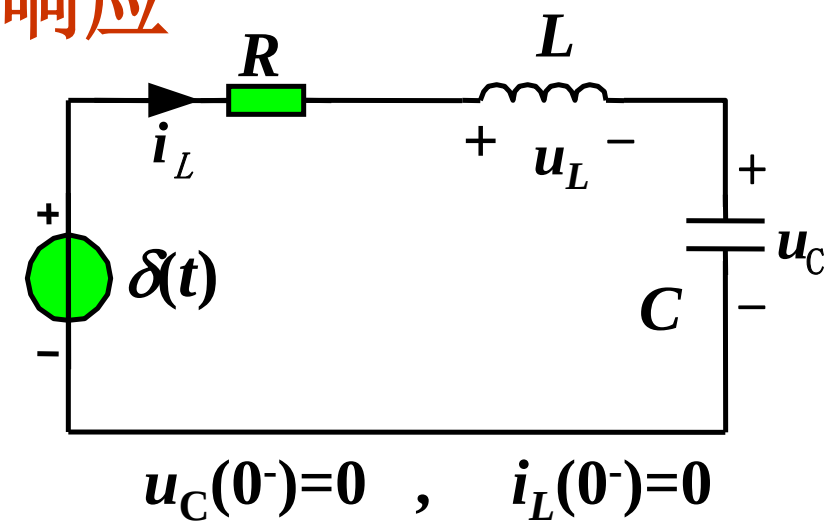
作业

- 初值：7-2，3
- 奇异电路初值：4*，5*
- 一阶电路：7-6，7，12，13，14，15，
- 二阶：7.18，20，22，24
- 16（冲激）

讨论：7-17，27，28

含半导体器件：7-29-36，29，31，33，35，36

§7.5.4 二阶电路的冲激响应



t 在0^-至0^+间

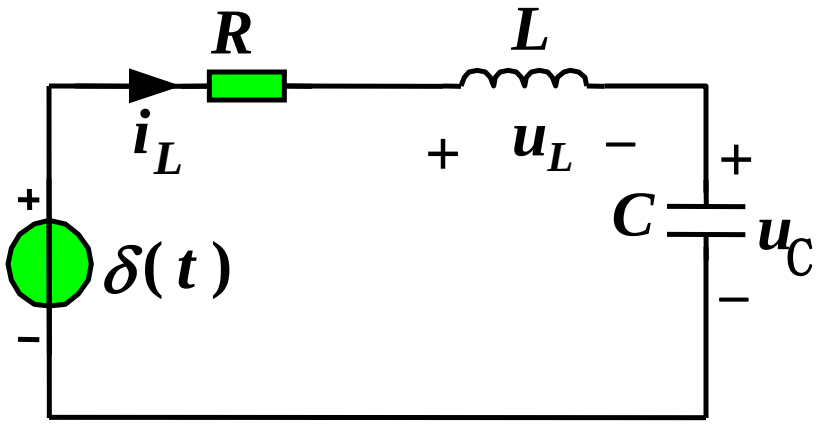
$$u_L = \delta(t)$$

$$i_L(0^+) = i_L(0^-) + \frac{1}{L}\int_{0^-}^{0^+} u_L dt = \frac{1}{L}$$

$$u_c(0^+) = u_c(0^-) = 0$$

$t > 0^+$ 为零输入响应

t 在0^-至0^+间



$$LC\frac{\mathrm{d}^2u_C}{\mathrm{d}t^2}+RC\frac{\mathrm{d}u_C}{\mathrm{d}t}+u_C=\delta(t)$$

$u_C(0^-)=0$ ， $i_L(0^-)=0$

u_C是跳变和冲激上式都不满足

设u_C不跳变，$\mathrm{d}u_C/\mathrm{d}t$ 发生跳变

$$\int_{0^-}^{0^+}LC\frac{\mathrm{d}^2u_C}{\mathrm{d}t^2}\mathrm{d}t+\int_{0^-}^{0^+}RC\frac{\mathrm{d}u_C}{\mathrm{d}t}\mathrm{d}t+\int_{0^-}^{0^+}u_C\mathrm{d}t=\int_{0^-}^{0^+}\delta(t)\mathrm{d}t$$

有限值

$$LC[\frac{\mathrm{d}u_C}{\mathrm{d}t}\Big|_{0^+}-\frac{\mathrm{d}u_C}{\mathrm{d}t}\Big|_{0^-}]+RC[u_C(0^+)-u_C(0^-)]=1$$

0

相等

$$LC\left[\frac{\mathrm{d}u_C}{\mathrm{d}t}\Big|_{0^+}-\frac{\mathrm{d}u_C}{\mathrm{d}t}\Big|_{0^-}\right]+RC[u_C(0^+)-u_C(0^-)]=1$$

$$LC\frac{\mathrm{d}u_C}{\mathrm{d}t}\Big|_{0^+}=1 \quad\Longrightarrow\quad C\frac{\mathrm{d}u_C}{\mathrm{d}t}\Big|_{0^+}=i_L(0^+)=\frac{1}{L}$$ 电感电流跳变

结论 $u_C(0^+)=u_C(0^-)=0$

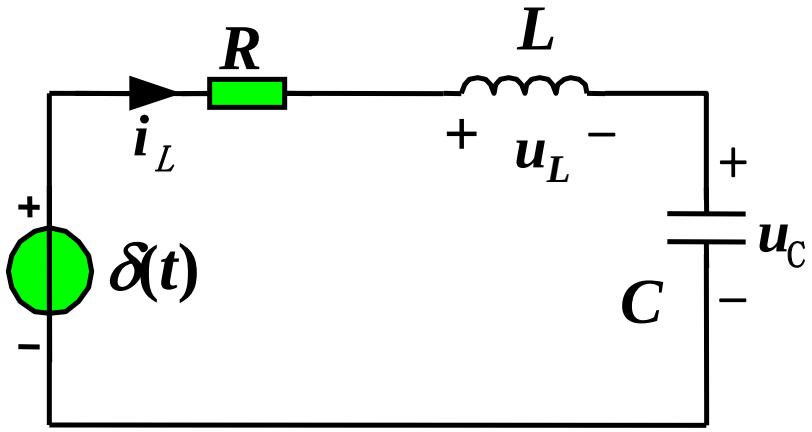
$$i_L(0^+)=\frac{1}{L}\neq i_L(0^-)$$

$$LC\frac{\mathrm{d}^2u_C}{\mathrm{d}t^2}+RC\frac{\mathrm{d}u_C}{\mathrm{d}t}+u_C=\delta(t)$$

对于RLC串联电路，在t＝0瞬时，电源电压全部加在电感上。

RLC串联时，冲击电压分配：电感上大于电阻大于电容。

$t>0^+$ 为零输入响应



$$LC\frac{d^2u_c}{dt^2}+RC\frac{du_c}{dt}+u_c=0$$

特征方程 $$p^2+\frac{R}{L}p+\frac{1}{LC}=0$$

$$(\frac{R}{L})^2-4\frac{1}{LC}\geq 0 \quad 即 \quad R\geq 2\sqrt{\frac{L}{C}} \qquad u_c=A_1e^{p_1t}+A_2e^{p_2t}$$

$$(\frac{R}{L})^2-4\frac{1}{LC}=0 \quad 即 \quad R=2\sqrt{\frac{L}{C}} \qquad u_c=(A_1+A_2t)e^{pt}$$

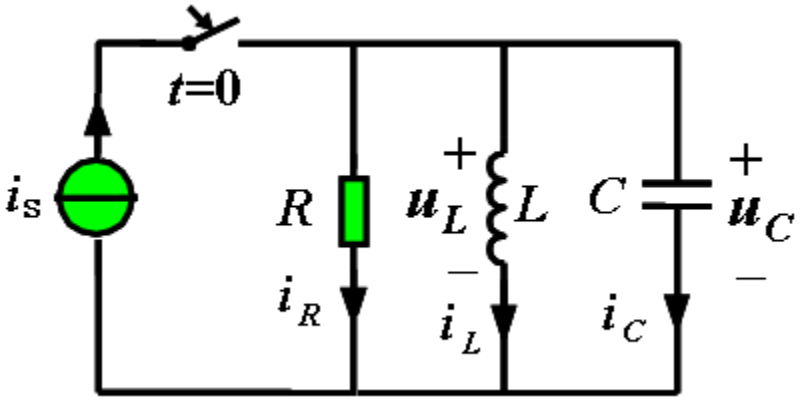
$$(\frac{R}{L})^2-4\frac{1}{LC}<0 \quad 即 \quad R<2\sqrt{\frac{L}{C}} \qquad u_c=Ke^{-\delta t}\sin(\omega t+\beta)$$

由初始值 $u_C(0^+)=u_C(0^-)=0$

$$i_L(0^+)=\frac{1}{L}\neq i_L(0^-)$$

定常数A_1, A_2 或 K, β

$$i_s = \delta(t)$$



$$C\frac{\mathrm{d}u_C}{\mathrm{d}t} + \int \frac{u_C}{L}\mathrm{d}t + \frac{u_C}{R} = \delta(t)$$

$$u_C(0^+) = \frac{1}{C} \neq u_C(0^-)$$

$$i_L(0^+) = i_L(0^-) = 0$$

对于RLC并联电路，在t＝0时，电容相当于瞬时短路。

RLC并联时，对于冲击电流，其流过的通路：电容易于电阻易于电感。